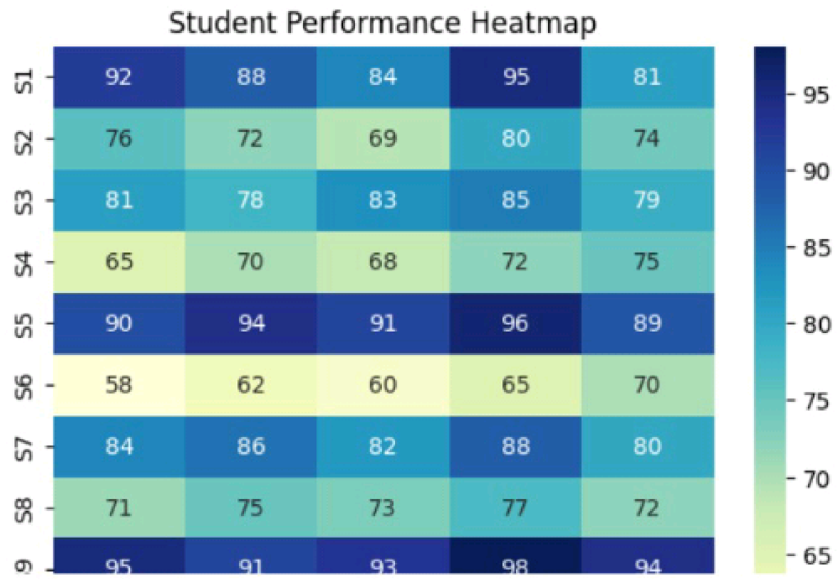
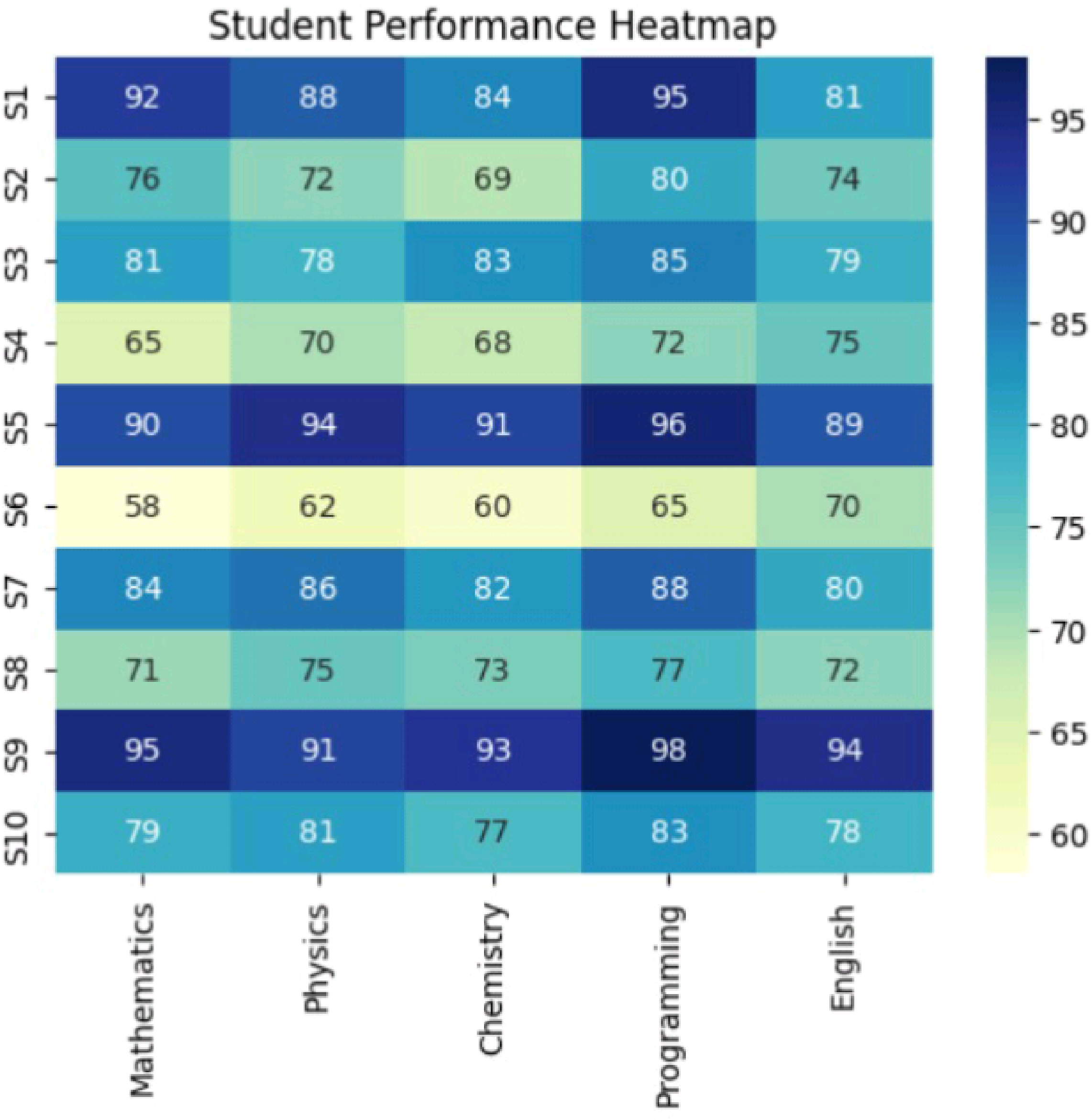
[2]

```python
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
data = {
    'Mathematics':[92,76,81,65,90,58,84,71,95,79],
    'Physics':[88,72,78,70,94,62,86,75,91,81],
    'Chemistry':[84,69,83,68,91,60,82,73,93,77],
    'Programming':[95,80,85,72,96,65,88,77,98,83],
    'English':[81,74,79,75,89,70,80,72,94,78]
}

students=['S1','S2','S3','S4','S5','S6','S7','S8','S9','S10']

df=pd.DataFrame(data,index=students)
sns.heatmap(df,annot=True,cmap='YlGnBu')
plt.title("Student Performance Heatmap")
plt.show()
```

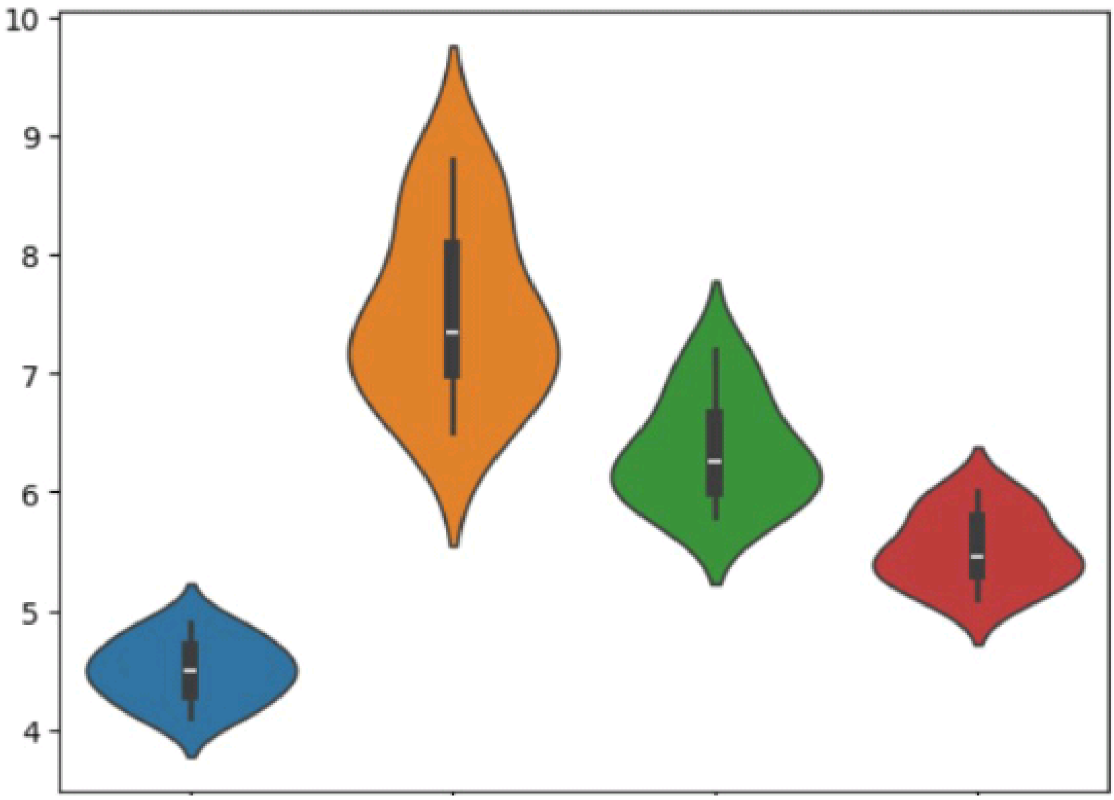


```
plt.title("Student Performance Heatmap")
plt.show()
```

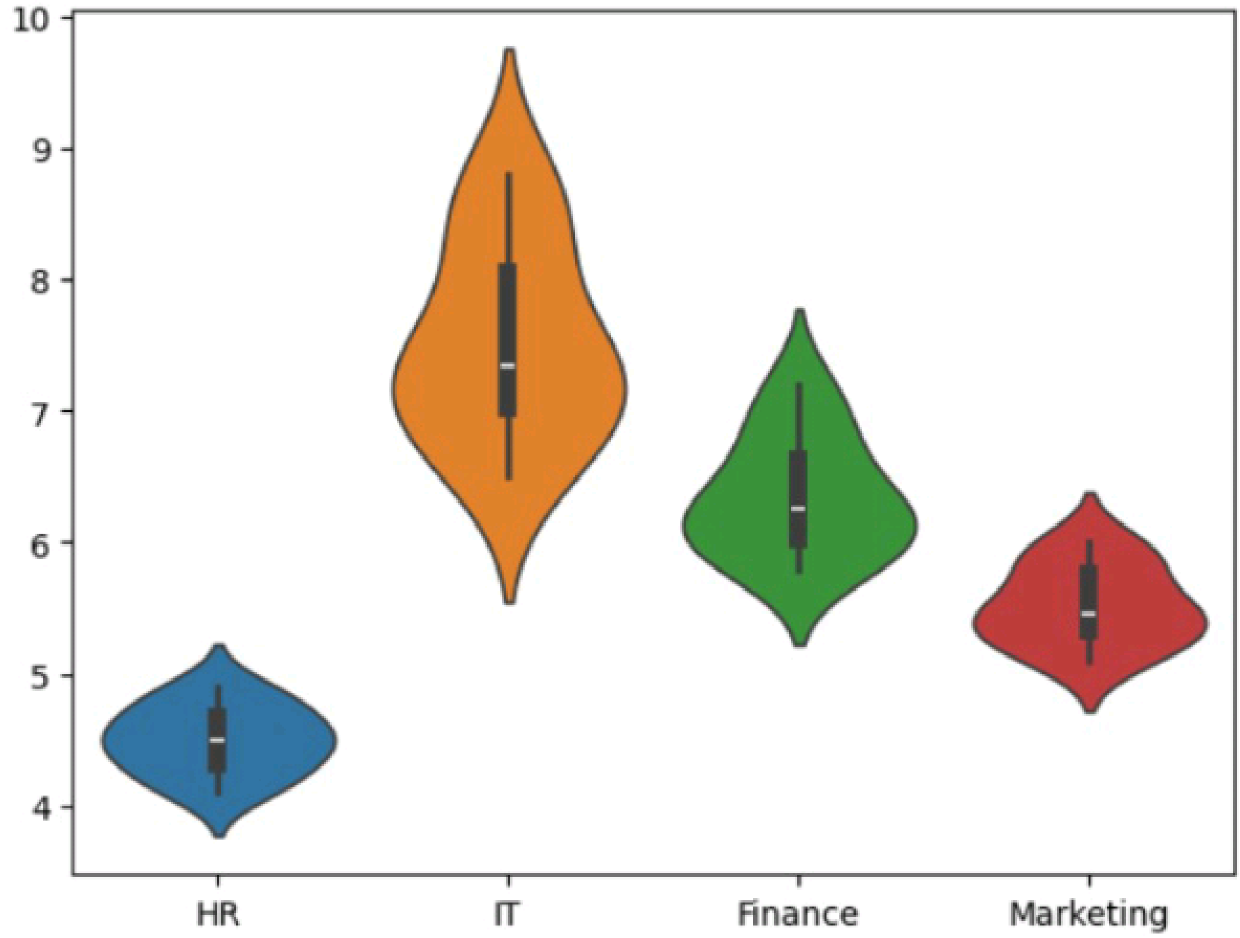


```python
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
data={
'HR':[4.2,4.5,4.7,4.1,4.6,4.3,4.8,4.4,4.9,4.5],
'IT':[6.5,7.0,7.4,6.8,8.2,7.6,8.5,7.1,8.8,7.3],
'Finance':[5.8,6.2,6.0,5.9,6.7,6.4,6.9,6.1,7.2,6.3],
'Marketing':[5.1,5.3,5.5,5.4,5.8,5.6,5.9,5.2,6.0,5.4]
}

df=pd.DataFrame(data)
sns.violinplot(data=df)
plt.show()
sns.boxplot(data=df)
plt.show()
```

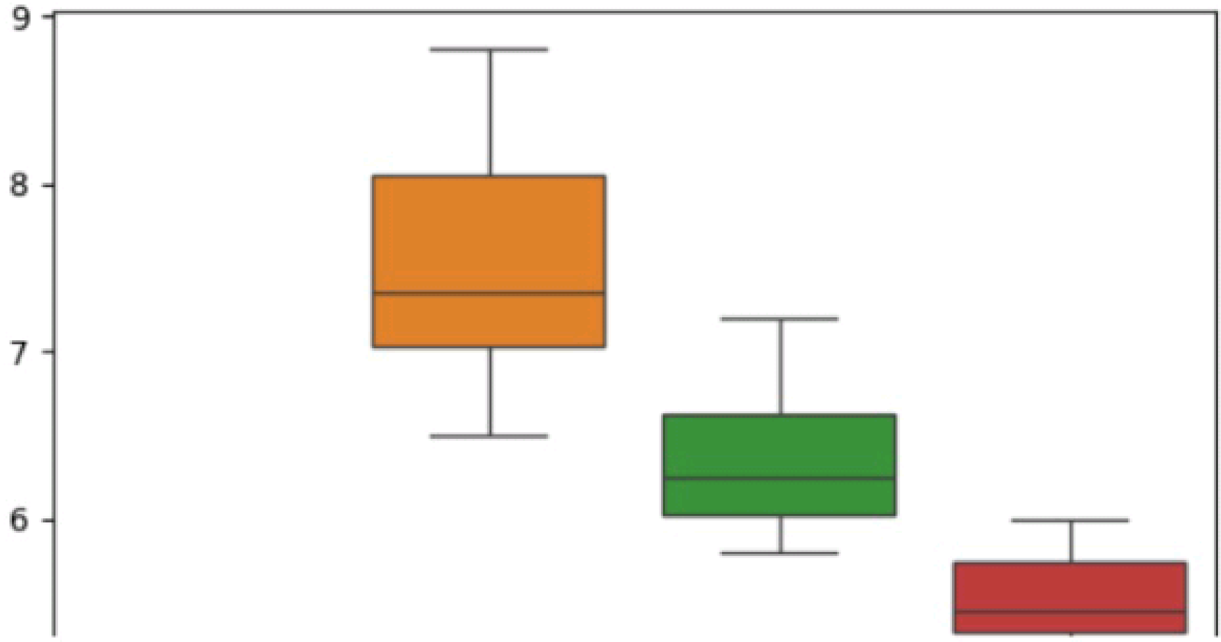
...



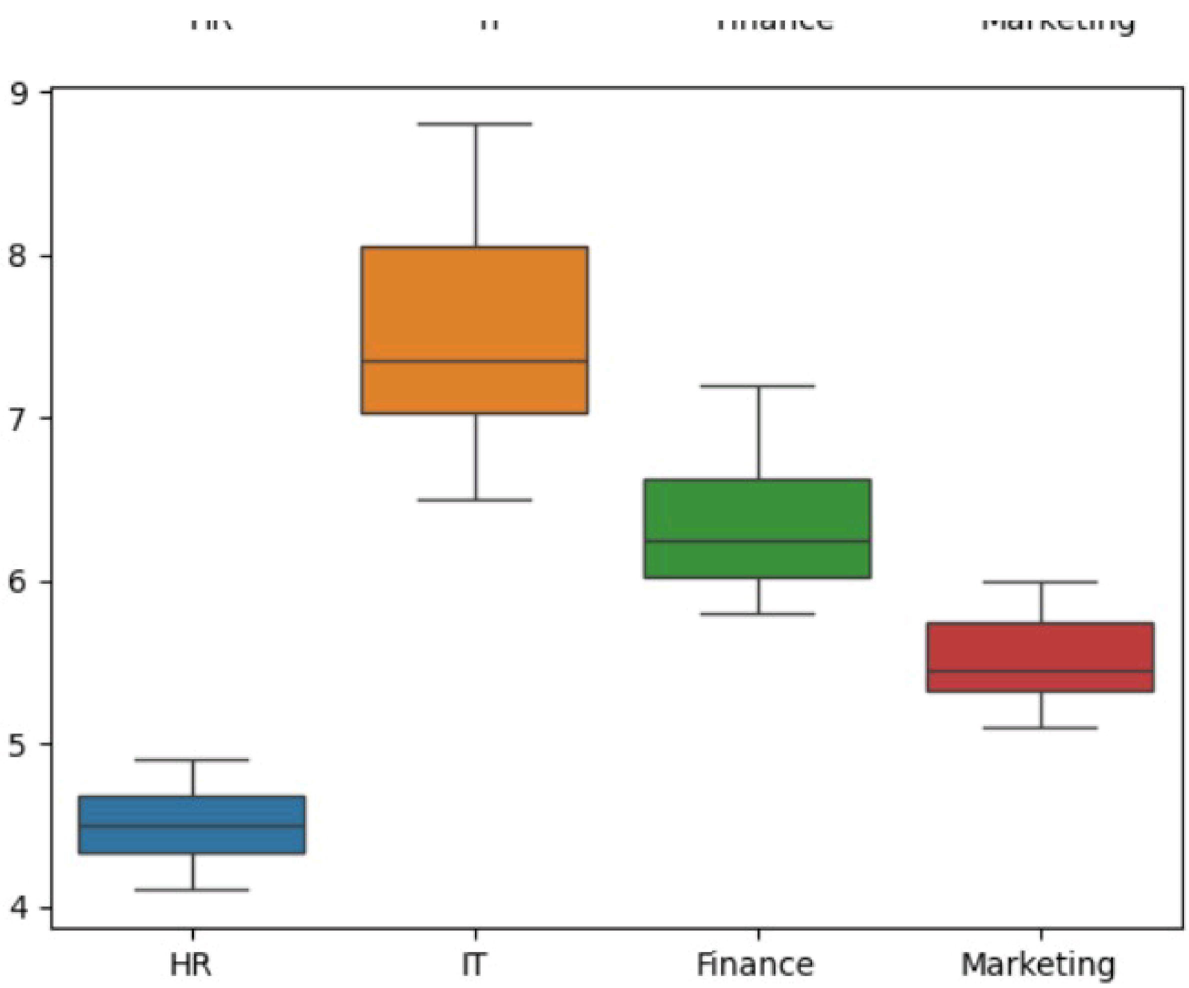
...



10
9
8
7
6
5
4
HR
IT
Finance
Marketing

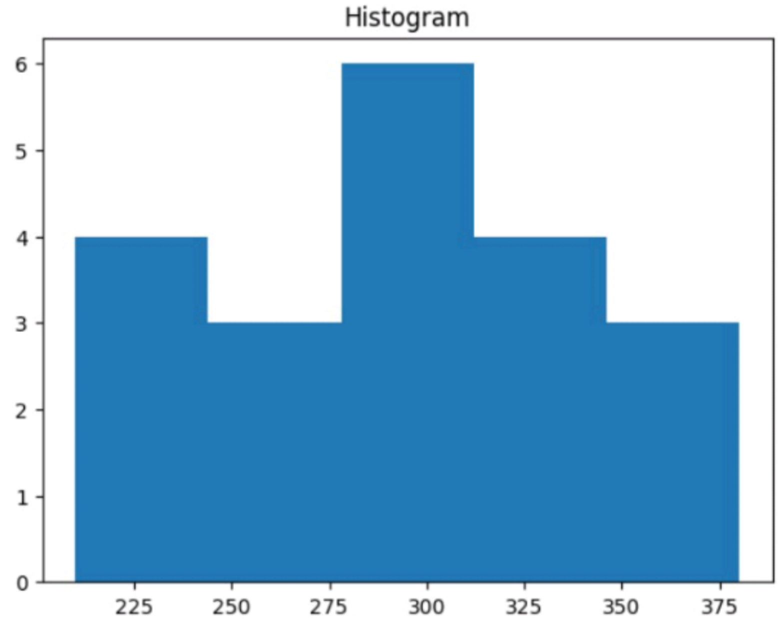


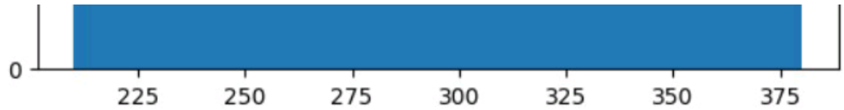
9
8
7
6

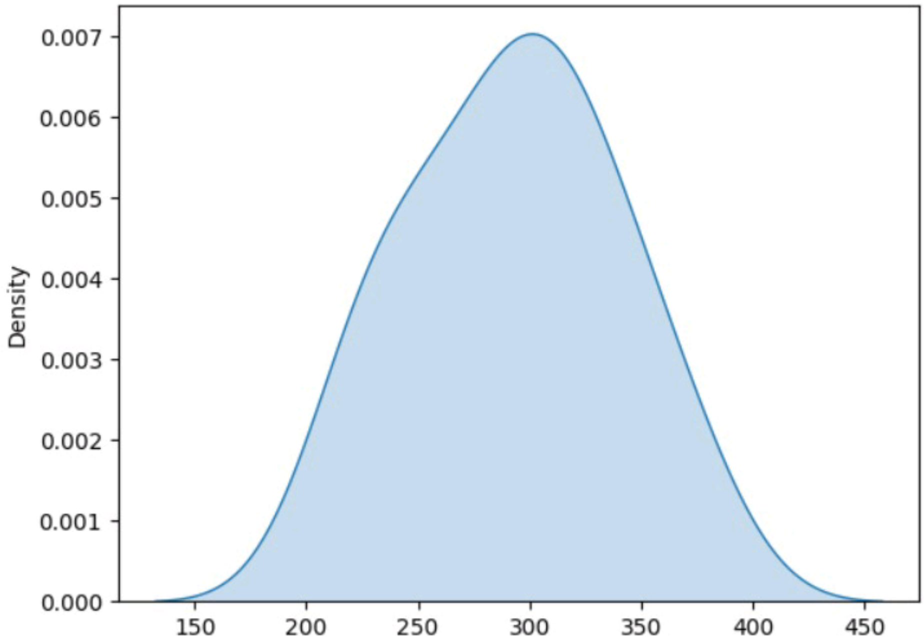

9
8
7
6
5
4
HR
IT
Finance
Marketing

[4]
0s

```python
import seaborn as sns
import matplotlib.pyplot as plt
response=[210,225,230,240,250,260,270,280,285,290,300,305,310,315,320,330,340,350,360,380]
plt.hist(response,bins=5)
plt.title("Histogram")
plt.show()
sns.kdeplot(response,fill=True)
plt.show()
```

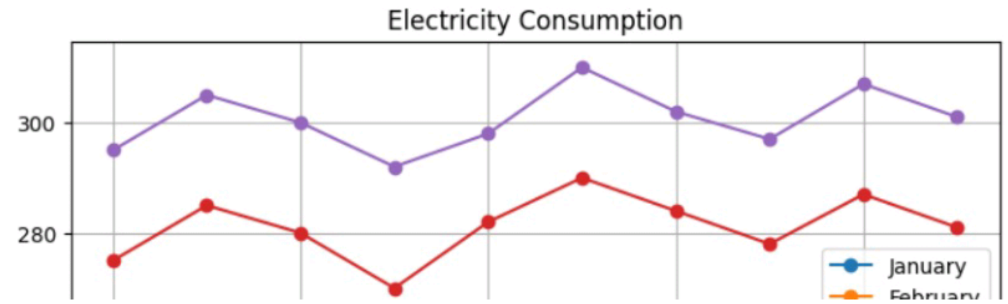



0
225
250
275
300
325
350
375

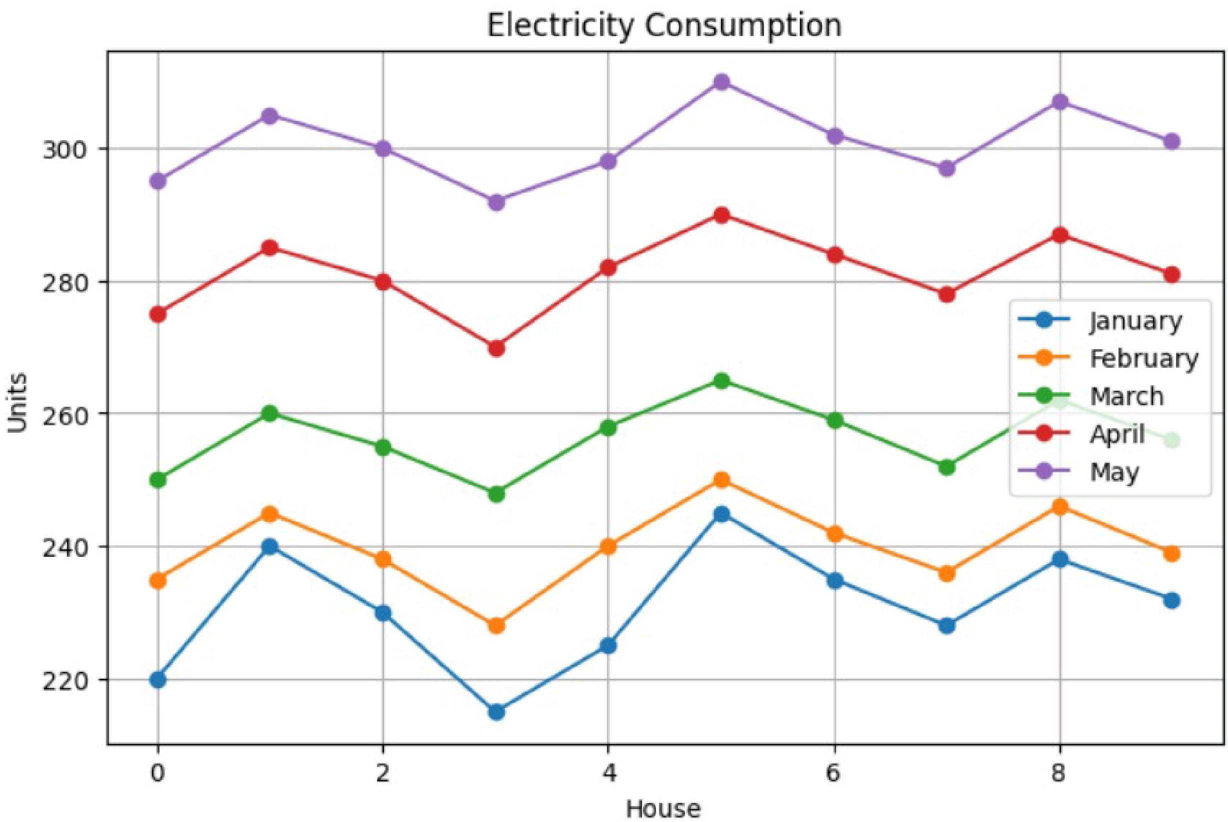

Density
0.000
0.001
0.002
0.003
0.004
0.005
0.006
0.007
150
200
250
300
350
400
450

10]
✓ 0s

```python
import pandas as pd
import matplotlib.pyplot as plt

data = {
    "January":[220,240,230,215,225,245,235,228,238,232],
    "February":[235,245,238,228,240,250,242,236,246,239],
    "March":[250,260,255,248,258,265,259,252,262,256],
    "April":[275,285,280,270,282,290,284,278,287,281],
    "May":[295,305,300,292,298,310,302,297,307,301]
}

df = pd.DataFrame(data)

df.plot(kind="line", marker="o", figsize=(8,5))
plt.title("Electricity Consumption")
plt.xlabel("House")
plt.ylabel("Units")
plt.grid(True)
plt.show()
```



...

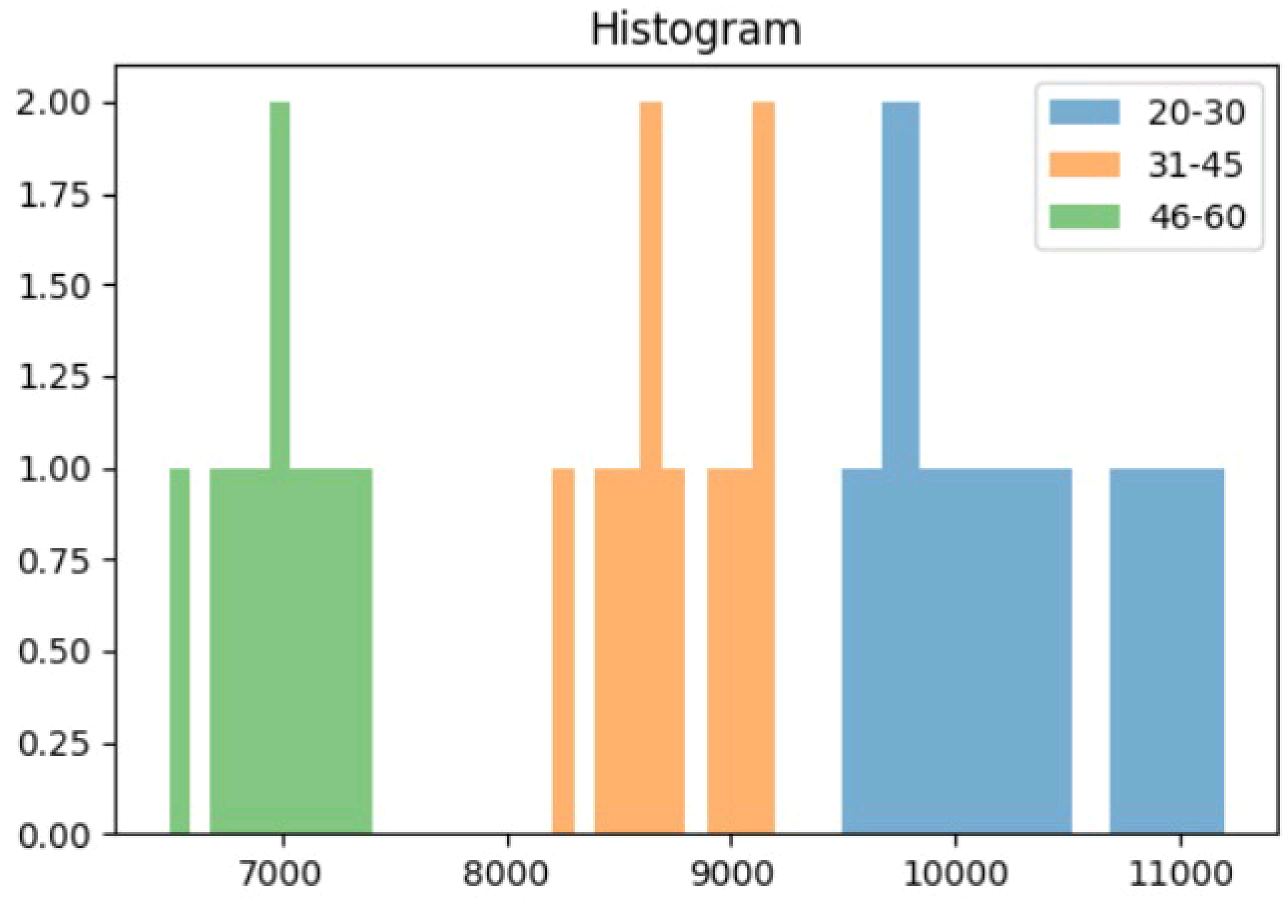


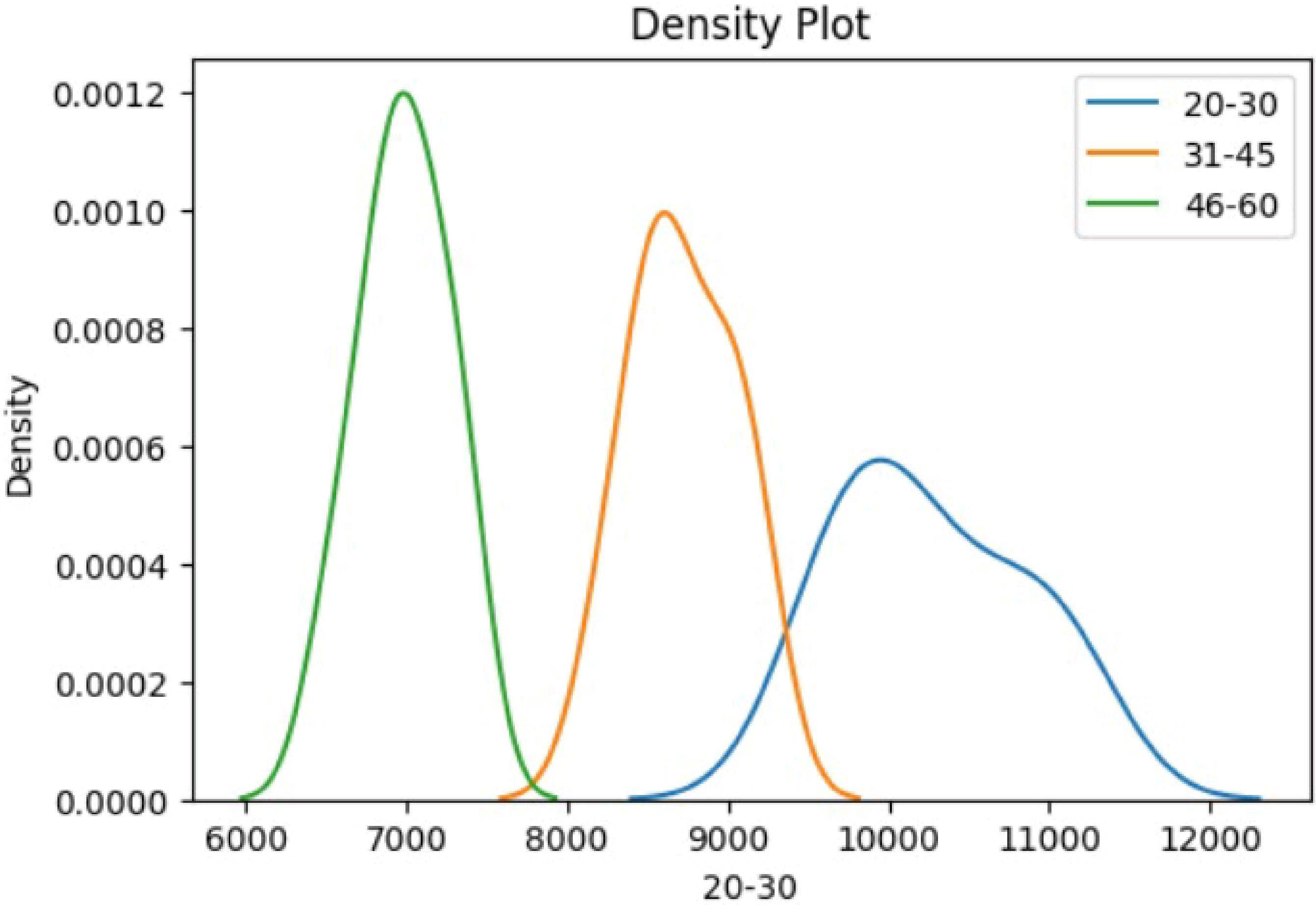
Electricity Consumption
300
280
260
240
220
Units
0
2
4
6
8
House
January
February
March
April
May

[8]
✓ 0s

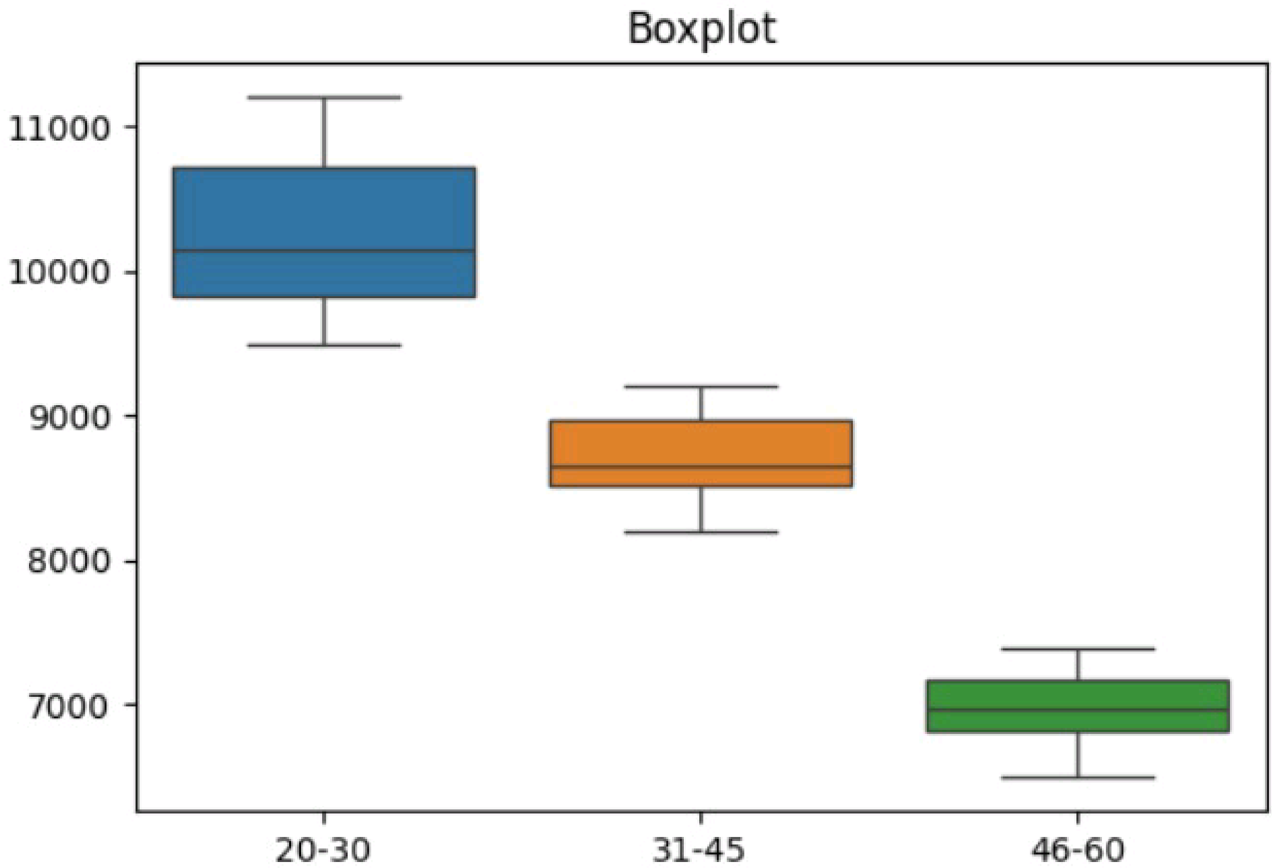
```python
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

data = {
    '20-30':[9500,10200,9800,11000,10500,9700,10100,10800,11200,9900],
    '31-45':[8200,8600,8400,9000,8900,8500,8700,9100,9200,8600],
    '46-60':[6500,6700,6900,7200,7000,6800,7100,7300,7400,6950]
}

df = pd.DataFrame(data)

# Histogram
plt.figure(figsize=(6,4))
plt.hist(df['20-30'], alpha=0.6, label='20-30')
plt.hist(df['31-45'], alpha=0.6, label='31-45')
plt.hist(df['46-60'], alpha=0.6, label='46-60')
plt.legend()
plt.title("Histogram")
plt.show()

# Density Plot
plt.figure(figsize=(6,4))
sns.kdeplot(df['20-30'], label='20-30')
sns.kdeplot(df['31-45'], label='31-45')
sns.kdeplot(df['46-60'], label='46-60')
plt.legend()
plt.title("Density Plot")
plt.show()

# Boxplot
plt.figure(figsize=(6,4))
sns.boxplot(data=df)
plt.title("Boxplot")
plt.show()
```

...


Histogram
20-30
31-45
46-60
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
7000
8000
9000
10000
11000



Density Plot
0.0012
0.0010
0.0008
0.0006
0.0004
0.0002
0.0000
Density
6000
7000
8000
9000
10000
11000
12000
20-30
20-30
31-45
46-60


Boxplot
11000
10000
9000
8000
7000
20-30
31-45
46-60